

2020 Physics H2 P2 Mark Scheme

1a	Effective resistance = $\frac{R_0}{2} = \frac{390}{2} = 195 \Omega$ <i>[For N identical resistors of resistance R in parallel, effective resistance is simply R/N]</i>	A1
1b	actual uncertainty for each resistor = 5% of $R_0 = 19.5 = 20 \Omega$ (to 1 SF) Since effective resistance $R = \frac{R_0}{2}$ $\Delta R = \frac{1}{2} \Delta R_0$ $= \frac{1}{2} (20)$ $= 10 \Omega$ (to 1 SF) Alternative method accepted: actual uncertainty for each resistor = 5% of $R_0 = 19.5 = 20 \Omega$ (to 1 SF) range of possible values for each resistor = $(390 \pm 20) \Omega$ i.e. 370Ω to 410Ω Maximum possible resistance = $\frac{410}{2} = 205 \Omega$ Minimum possible resistance = $\frac{370}{2} = 185 \Omega$ Total $\Delta R = \frac{\text{max} - \text{min}}{2} = \frac{205 - 185}{2} = 10 \Omega$	C1 A1 C1 A1
1c	$R = V/I = 194.1935$ $\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$ $\frac{\Delta R}{194.19} = \frac{0.01}{6.02} + \frac{0.001}{0.031}$ $\Delta R = 7 \Omega$	C1 C1 A1